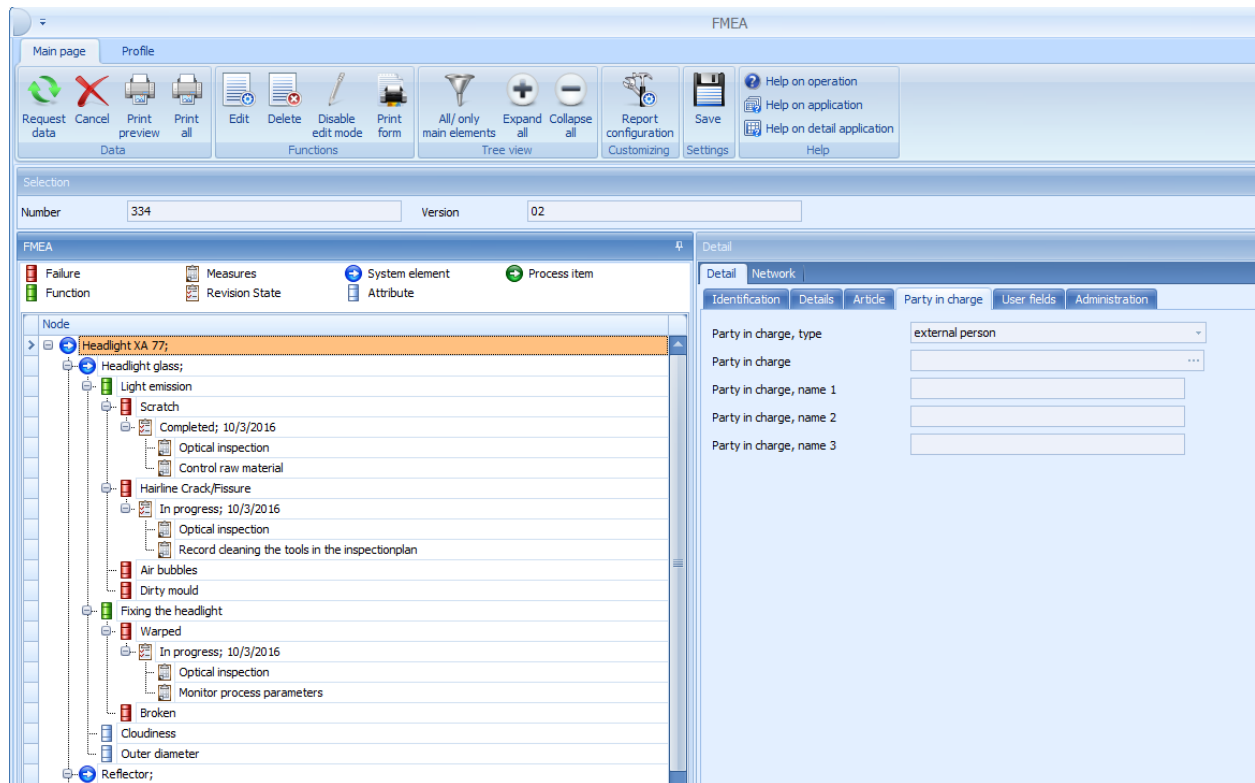


# 1 FMEA (Failure Mode and Effects Analysis)

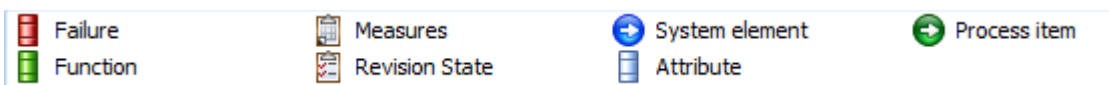
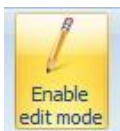
|                        |                                                                                                                         |
|------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Menu                   | Quality management → FMEA → FMEA ->  |
| Transaction code       | fmea                                                                                                                    |
| Function authorization | fmea, fmevnu, fmevnue, tmg, fmeastru, fmeadoc, failnet, doctype, DOC-LINK, docli, fmeaexport, fmeacpl                   |

| Available user fields                                                                               |                            |               |
|-----------------------------------------------------------------------------------------------------|----------------------------|---------------|
| Where                                                                                               | Object type/user field key | Source (type) |
| Element type system<br>Detail view                                                                  | FMEASTRUCTURE/SYSTEMELEM   | MF-D          |
| Element type process<br>Detail view                                                                 | FMEASTRUCTURE/PROCESS      | MF-D          |
| Element type function<br>Detail view                                                                | FMEASTRUCTURE/FUNCTION     | MF-D          |
| Element type failure<br>Detail view                                                                 | FMEASTRUCTURE/FAILURE      | MF-D          |
| Element type characteristic<br>Detail view                                                          | FMEASTRUCTURE/SYSTEMELEM   | MF-D          |
| Element type revision state<br>Detail view                                                          | FMEASTRUCTURE/MEASURELEVEL | MF-D          |
| Element type measure<br>Detail view                                                                 | FMEASTRUCTURE/MEASURE      | MF-D          |
| <a href="#">How to configure user fields?</a> <a href="#">Which user field types are available?</a> |                            |               |



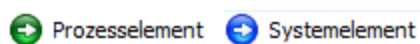
## Field descriptions

The following paragraphs list the available selection fields. To insert the fields into the tree, you can use drag and drop. (Edit mode must be activated)



The symbol of the mouse pointer reveals if you can insert the element selected below the element where the mouse pointer is placed.

## Field description



Enter new process/system element using drag and drop. (Edit mode must be activated). To insert the selected element, release the mouse key. The *Insert* dialog opens automatically.

**Tab Identification****Element type**

The element type is either "Process element" or "System element".

**Designation (name)**

You can enter any description into the field

**Identification**

Alphanumerical field for the element number, for example 1.4.1 for the first entry in level 3. This field is maintained manually and can remain empty.

**Comment**

You can add further information in here.

**Tab Details****operation**

Number of the operation including selection list. Can be entered manually. Operation can be non existent.

**Operation name**

Name of the operation. Can be entered manually.

**Machine**

The workplace catalog is available for selection. By default, the list is filtered for resources of the type MNR (machine number). You can change the filter. When confirming the selection, the workplace number is transferred.

**Machine name**

Display of the name of the assigned machine.

**Machine group**

The workplace catalog is available for selection. By default, the list is filtered for resources of the type MNR (machine number). You can change the filter. When confirming the selection, the system transfers the selected workplace group. If you assign the machines directly, the system does not automatically transfer the group of the selected machines.

**Revision state**

You can manually enter a date which identifies the revision state of this process element.

**Tab *Article*****Article number**

The article list is available for selection. After opening the list, the system filters only active articles. After confirming the selection, the system transfers the fields "Article number", "Drawing issue no" and "Article name".

**Article name/designation**

The system dynamically identifies and displays the article name for the article number and drawing issue number displayed.

**Drawing issue number**

Drawing issue number of displayed article

**Tab *Party in charge*****Party in charge, type**

Selection list including the long status names of the active statuses of status type *Party in charge*. If you create a new party in charge, the long name of the status "Person" is displayed by default.

**Party in charge**

Content of the field "Number" of the selected data record, e.g. number of the selected person. Directly after opening, the system only displays active data records from the selected type. After selecting, the contents of the fields "Name 1", "Name 2" and "Name 3" are accepted and then transferred into the respective name fields.

**Party in charge, Name 1, Name 2, Name 3**

The field contents of Name 1, Name 2 and Name 3 of the party in charge are shown. The customer name and the content of the address fields 1 and 2 are displayed for customers. The last name, first name and initials are displayed for external persons.

 Failure

| Identification      | User fields                                                                                                          | Administration |
|---------------------|----------------------------------------------------------------------------------------------------------------------|----------------|
| Identification      | <input type="text"/>                                                                                                 |                |
| Element type        | Failure                                                                                                              |                |
| Failure             | 12-001                                                                                                               |                |
| Failure designation | Forms no beam                                                                                                        |                |
| Comment             | <input type="text"/>                                                                                                 |                |
| Evaluation S        |                                                                                                                      |                |
| Evaluation S        | 5                                                                                                                    |                |
| Evaluation name     | moderate                                                                                                             |                |
| Description         | Operability of the vehicle restricted, immediate workshop stay required. Failure of key control and comfort systems. |                |

### Tab Identification

#### Element type

The element type is either "Process element" or "System element".

#### Failure

Selection of the failure type from the master data catalog. Select a failure. The failure name is then added in the field *Failure designation*. You can change or extend the name, if required. After the initial entry of a failure number, the system checks if this failure exists in the master data catalog on saving the entry. If not, an error message occurs.

#### Failure designation

The failure name is added to the master data entry of the previously selected failure number.

#### Identification

Alphanumeric field for the element number, for example 1.4.1 for the first entry in level 3. This field is maintained manually and can remain empty.

#### Comment

You can add further information in here.

### Tab FMEA Rating numbers

#### Evaluation S

Requests list of the detail application FMEA rating numbers. The list is filtered by the entries of the evaluation catalog assigned in the FMEA header and of the evaluation type "S". The rating number is taken over.

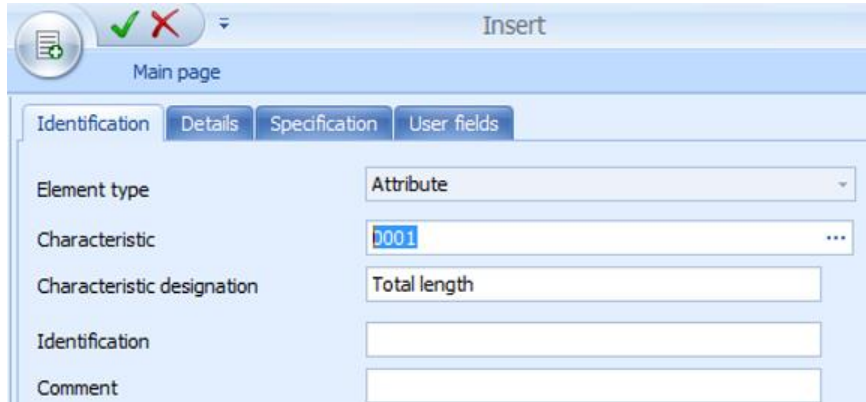
#### Evaluation name

Display of the evaluation name relating to the rating number previously selected.

## Description

Display of the description of the rating number previously selected.

 Attribute



### Tab Identification

#### Element type

The element type is either "Process element" or "System element".

#### Characteristic

Select a characteristic from the master data catalog. After selecting, the failure name is integrated into the characteristic name field and can be changed or extended if required. When you directly enter a characteristic number, the system checks if the characteristic is available in the master data catalog. If not, an error message occurs.

#### Characteristic designation

The characteristic name is added to the master data entry of the characteristic number previously selected.

#### Identification

Alphanumerical field for the element number, for example 1.4.1 for the first entry in level 3. This field is maintained manually and can remain empty.

#### Comment

You can add further information in here.

## Tab Details

The screenshot shows the 'Details' tab of a software interface. It contains several sections with configuration options:

- Classification:** A dropdown menu.
- FMEA form:** A checked checkbox.
- Inspection plan:** A checked checkbox.
- Production control plan:** An unchecked checkbox.
- Characteristic type:** Three radio buttons: 'attributive' (selected), 'inspection hint', and 'variable'.
- Inspection result base:** Two radio buttons: 'all samples' (selected) and 'last sample'.
- Gage type:** Two radio buttons: 'Gage number' (selected) and 'Gage group'.
- Gage:** A text field containing '08-01-01'.
- Gage designation:** An empty text field.

### Classification

This function ranks a critical or crucial characteristic. For information purposes only.

### FMEA form

Specifies if the given characteristic is a part of the FMEA form.

### Inspection plan

Specifies if the given characteristic is a part of the inspection plan. Only for information purposes.

### Characteristic type

This option specifies whether the collection of measured values (variable) or the identification of the number of detected failures (attributive) is used for the inspection. In case of an attributive inspection, the decision is often only based on the "pass" or "fail" results. Further characteristic types are the inspection chart and the information characteristic. The information characteristic is only used to display a document during the inspection process. Subject to the input type, the lower area of the dialog provides the respective sampling schemes.

### Inspection result base

This setting defines whether all samples or only the sample recorded last is used to identify the inspection result (pass/fail).

### Gage type

This defines if a test equipment or a test equipment group should be assigned to a characteristic.

### Test equipment (gage)

This field is not visible if the *Gage type* "Test equipment group" is selected. The resource catalog is available for selection. The system filters by resource type PRM (test equipment). When confirming the selection, the resource and the resource name is accepted in the fields "Gage" and "Gage designation".

### Gage designation (name of test equipment)

After selecting the test equipment, the system displays the name. (This field is invisible if the *inspection plan type* "Test equipment group" is selected.)

### Test equipment (gage) group

This field is not visible if the *test equipment type* "Test equipment number" is selected. The resource family catalog is available for selection. The system filters by resource type PRM (test equipment). When confirming the selection, the resource family and the resource family name is accepted in the field "Group" and "Group name". If you select and insert a characteristic from the characteristic catalog, this field is populated with the respective contents of the selected characteristic.

### Group name (designation)

After selecting the test equipment group, the system displays the name. (This field is invisible if the *inspection plan type* "Test equipment number" is selected.)

### Tab Specification

| Field                                   | Value                    |
|-----------------------------------------|--------------------------|
| Sampling scheme                         | n-c inspection           |
| Sample size                             | 7                        |
| Acceptance qty.                         | 0                        |
| Decimal places                          | 5                        |
| Size                                    | absolut                  |
| Standard                                |                          |
| Fit                                     |                          |
| Calculate limit values based on the fit | <input type="checkbox"/> |
| Upper TL                                | 130,25                   |
| Target value                            | 130,00                   |
| Lower TL                                | 129,90                   |
| Unit                                    | mm                       |

### Sampling scheme

The sampling scheme defines the inspection procedure. In case of an n-c inspection and parameters 5-0, 5 pieces are checked and 0 failures may be detected.

### Sample size

If you select and insert a characteristic from the characteristic catalog, this field is populated with the respective contents of the selected characteristic.

### Acceptance quantity

Here you can define the number of authorized failed parts. This field is only visible if NC is selected during a sampling scheme. Initially assigned with "0". If you select and insert a characteristic from the characteristic catalog, this field is populated with the respective contents of the selected characteristic.

**Decimal places**

The specified number of decimal places is later displayed in the tree structure. If you select and insert a characteristic from the characteristic catalog, this field is populated with the respective contents of the selected characteristic.

**Size**

Tolerance limits can be stated in absolute, relative and proportional values. Please note that relative and proportional lower limits must be stated in negative numbers.

**Standard**

Calculation of tolerances based on specific standards (e.g. ISO metric fits). Subject to the selected standard, further information is requested (e.g. engineering fit). The system automatically calculates the tolerance limits on the basis of these specifications. Displays the active status entries. If you select and insert a characteristic from the characteristic catalog, this field is populated with the respective contents of the selected characteristic.

**Fit**

Calculation of tolerance limits on the basis of a specific standard and engineering fit. The selected fit depends on the selected standard. The system displays entries accordingly. The content changes due to the selected fit. The content of the selected characteristic is assigned to the field when selecting and accepting a characteristic from a catalog.

**Calculating tolerance limits with fits**

If you activate this field, tolerance limits are automatically calculated according to the standard and requirement for fits.

**Upper TL/Lower TL**

Enter/display the upper and lower tolerance limit. If you select and insert a characteristic from the characteristic catalog, this field is populated with the respective contents of the selected characteristic. (Only visible if the field "Size" *absolute* was selected).

**Upper TL rel. % and lower TL in %**

Enter/display the upper and lower tolerance limit. If you select and insert a characteristic from the characteristic catalog, this field is populated with the respective contents of the selected characteristic. (Only visible if the size 'relative' was chosen).

**Target value**

Entry/display of the target value. If you select and insert a characteristic from the characteristic catalog, this field is populated with the respective contents of the selected characteristic.

**Unit**

The unit catalog is available for selection. The system accepts the unit after confirming the selection. If you select and insert a characteristic from the characteristic catalog, this field is populated with the respective contents of the selected characteristic.

 Function

| Identification | User fields          | Administration |
|----------------|----------------------|----------------|
| Identification |                      |                |
| Element type   | Function             |                |
| Function type  | Elementary           |                |
| Comment        |                      |                |
| Designation    | Fixing the headlight |                |

**Tab Identification****Element type**

The element type is either "Process element" or "System element".

**Function type**

ID of the function type

**Designation (name)**

Any description of the function element can be entered.

**Identification**

Alphanumeric field for the element number, for example 1.4.1 for the first entry in level 3. This field is maintained manually and can remain empty.

**Comment**

You can add further information in here.

## Revision State

| Identification       |                                                                                                                                                                                                                                                                          | Details | User fields | Administration |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------------|----------------|
| Identification       | <input type="text"/>                                                                                                                                                                                                                                                     |         |             |                |
| Element type         | Revision State                                                                                                                                                                                                                                                           |         |             |                |
| Comment              | <input type="text"/>                                                                                                                                                                                                                                                     |         |             |                |
| Evaluation O         |                                                                                                                                                                                                                                                                          |         |             |                |
| Evaluation O         | 5                                                                                                                                                                                                                                                                        |         |             |                |
| Evaluation name      | moderate                                                                                                                                                                                                                                                                 |         |             |                |
| Description          | New development of systems / components with experience from previous developments or detailed changes under comparable operating conditions. Proven system / components with many years of damage-free series production experience under changed operating conditions. |         |             |                |
| Evaluation D         |                                                                                                                                                                                                                                                                          |         |             |                |
| Evaluation D         | 1                                                                                                                                                                                                                                                                        |         |             |                |
| Evaluation name      | very high                                                                                                                                                                                                                                                                |         |             |                |
| Description          | Very high probability of detection of the malfunction by proven methods of verification of the previous generation. The effectiveness of the detection action has been demonstrated for this product.                                                                    |         |             |                |
| RPN                  |                                                                                                                                                                                                                                                                          |         |             |                |
| Risk priority number | 40                                                                                                                                                                                                                                                                       |         |             |                |

### Tab Identification

#### Element type

The element type is either "Process element" or "System element".

#### Identification

Alphanumerical field for the element number, for example 1.4.1 for the first entry in level 3. This field is maintained manually and can remain empty.

#### Comment

You can add further information in here.

#### Evaluation O (occurrence)

Selection of likelihood of occurrence as stated in the catalog of the FMEA overview.

#### Evaluation name

See catalog entry.

### Description

See catalog entry.

### Evaluation D (Detection)

Selection of likelihood of detection of a failure as stated in the catalog of the FMEA overview.

### Evaluation name

See catalog entry.

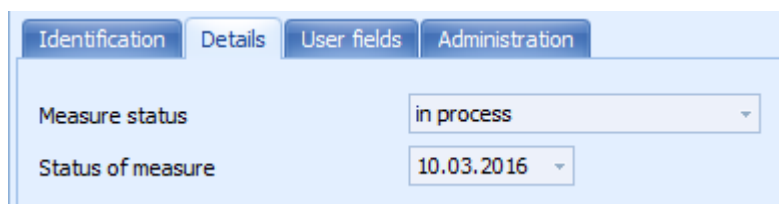
### Description

See catalog entry.

### Risk Priority Number (RPN)

Calculated by multiplying scores of likelihood of failure occurring or being detected.  $S \times O \times D$ . The value S is generated from the highest S-value of all possible failure modes. Rejected measures are not used in the RPN calculation. If no detection or avoidance measure is available, the RPN calculation is not conducted.

### Tab Details



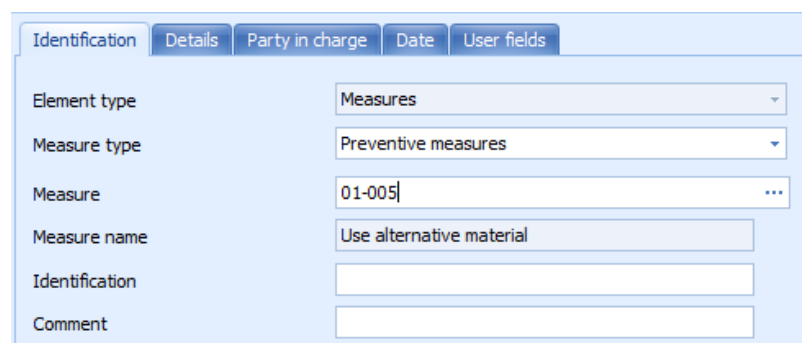
### Measure status

The measure status is generated from subordinate measures which shows the lowest processing status and displayed here. The measure status is displayed here.

### Revision state

Date when measures were taken.

### Measures



**Tab Identification****Element type**

Identifier of the element type, in this case "Measure element"

**Measure type**

You can define if it is an avoidance or a detection measure.

**Measure**

Selection of a measure from the master data catalog. Select a measure. The name is then used for the field *Measure name*. You can change or extend the field content, if required.

**Identification**

Description of the hierarchical element assignment within the overall tree structure using a "Numbering".

**Comment**

You can add further information in here.

**Tab Details**

The screenshot shows a software interface with five tabs: 'Identification', 'Details', 'Party in charge', 'Date', and 'User fields'. The 'Details' tab is selected. It contains the following fields:

- Measure text**: A text input field.
- Measure status**: A dropdown menu with 'finished' selected.
- Show externally**: A checkbox that is checked.
- Fulfillment in %**: A text input field with '100' entered.
- Effectiveness in %**: A text input field with '100' entered.

**Measure text**

You can add further information in here.

**Measure status**

The processing status is documented using the above status.

**Show externally**

This field is used for information only. This is the basis to control which measures should be shown when external people are present. We recommend to activate that field.

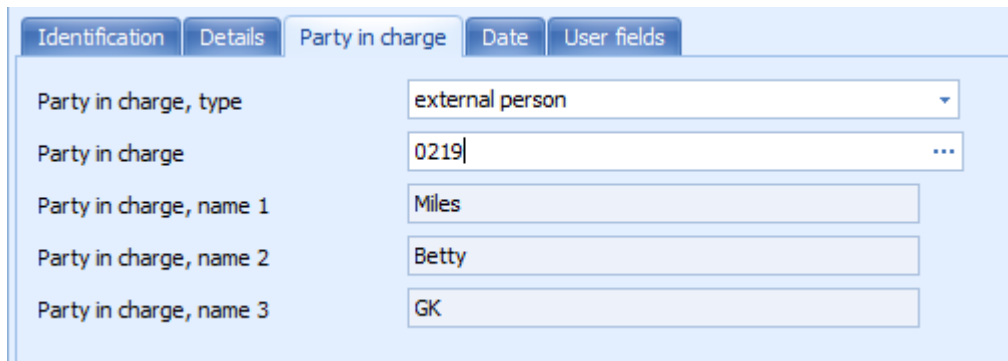
**Fulfillment level in %**

Fulfillment level is documented here. Numbers from 0 until 100 can be entered.

**Effectiveness in %**

Effectiveness is documented here. Numbers from 0 until 100 can be entered.

### Tab *Party in charge*



#### Party in charge, type

Selection list including the long status names of the active statuses of status type *Party in charge*. If you create a new party in charge, the long name of the status "Person" is displayed by default.

#### Party in charge

Content of the field "Number" of the selected data record, e.g. number of the selected person. Directly after opening, the system only displays active data records from the selected type. After selecting, the contents of the fields "Name 1", "Name 2" and "Name 3" are accepted and then transferred into the respective name fields.

#### Party in charge, Name 1, Name 2, Name 3

The field contents of Name 1, Name 2 and Name 3 of the party in charge are shown. The customer name and the content of the address fields 1 and 2 are displayed for customers. The last name, first name and initials are displayed for external persons.

### Tab *Date*

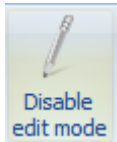
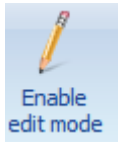


#### Target date

Measures must be completed or rejected until the target date.

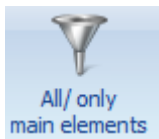
The actual date is entered manually. It documents when the measure was completed or rected.

## Toolbar



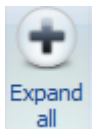
### Enable edit mode

To edit an FMEA, you must enable the edit mode. This mode blocks an opened FMEA for all other users. The FMEA is read-only. When editing is completed, disable the edit mode.



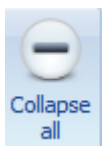
### All / only main elements

When using this button, the system shows the tree view filtered for main elements. Main elements are system and process elements. If you select this button again, you switch to view all elements ("all").



### Expand all

If you select this button, the tree view expands. If you use a filter, only main elements are expanded. If you select all elements (default setting), all elements expand.



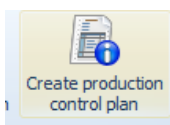
### Collapse all

If you select this button, the tree view collapses. The filter "all / only main elements" is not affected by it.



### Print form

This button prints an FMEA form for a selected process or system element.



### Calling the production control plan

Use this button to display and print a production control plan using the FMEA data. Requirement: You must select a system or process element and the "FMEA-PLP" license must be available.



#### Export of characteristic data

Use this button to display and print a production control plan using the FMEA data. Requirement: You must select a system or process element and the "FMEA-PLP" license must be available.

## Detail application "Production control plan"

|         |          |
|---------|----------|
| License | FMEA-PLP |
|---------|----------|

All characteristic elements are identified for the selected system/process element that are located below the element and have an active checkbox "Production control plan" in the "Details" tab. Characteristics are also included which belong to lower-level system and process elements.

In the printed form, characteristics belonging to a system element are printed in the "Product" column. Characteristics of a process element are displayed in the column "Process".

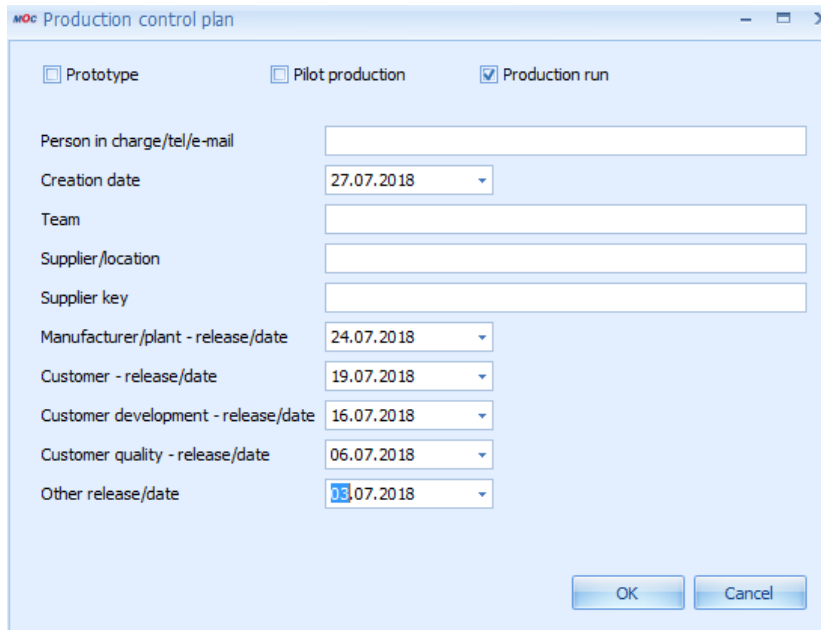
The part / process number and the process name are derived from the "Part number", "Operation" and "Operation designation" fields of the associated system / process element.

The "Inspection system" column displays the test equipment or test equipment group name and in brackets the associated identification number. The contents of the columns "Frequency", "Control method" and "Action if fail" are defined in the the following user fields of the FMEA characteristics:

- Frequency: userfield\_c63 (20 alpha numerical characters)
- Control method: userfield\_c65 (40 alpha numerical characters)
- Action with fail: userfield\_c66 (40 alpha numerical characters)

The customer creates the user fields.

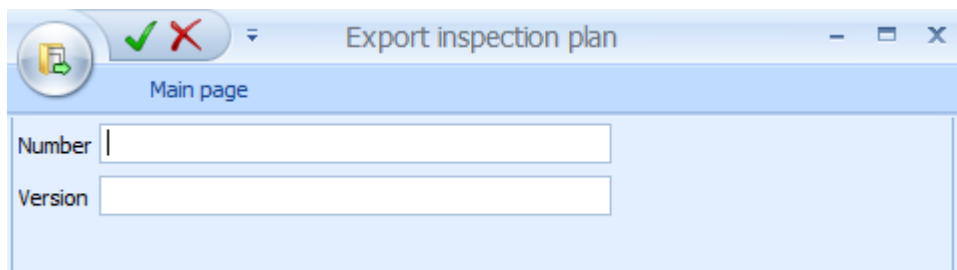
To define the header data of the production control plan, an input dialog is opened before the actual printout.



## Detail application "Export inspection plan"

|         |          |
|---------|----------|
| License | FMEA-PPL |
|---------|----------|

If you press the button "Export inspection plan", a window opens. Enter a number and a version. The number and version entered are used for the data file that is then created.



Confirm the entry. The system identifies all characteristic elements of the selected system/process element. The system only uses the elements, if two conditions are fulfilled: the characteristic elements are one level below the system/process element and for the respective characteristics, the option "Inspection plan" is enabled in tab "Details". The characteristics identified are exported into a CSV file. The CSV data file includes a header and a data row for each characteristic.

The file name is structured as follows:

FMEA\_<number>\_<version>\_<article number>\_<drawing issue number>\_<YYYY-MM-DD>-  
T<HH-MM-SS>.123csv

|                        |                                                                            |
|------------------------|----------------------------------------------------------------------------|
| FMEA:                  | always FMEA                                                                |
| <number>               | number of input dialog                                                     |
| <version>              | version of input dialog                                                    |
| <article number>       | article number of the selected process/system element                      |
| <drawing issue number> | drawing issue number of the article of the selected process/system element |
| <YYYY-MM-DD>           | system date at the point in time of export                                 |
| T                      | always T to make clear that the time is specified in the following.        |
| <HH-MM-SS>             | time of export                                                             |
| .123                   | number of milliseconds at the point in time of export                      |

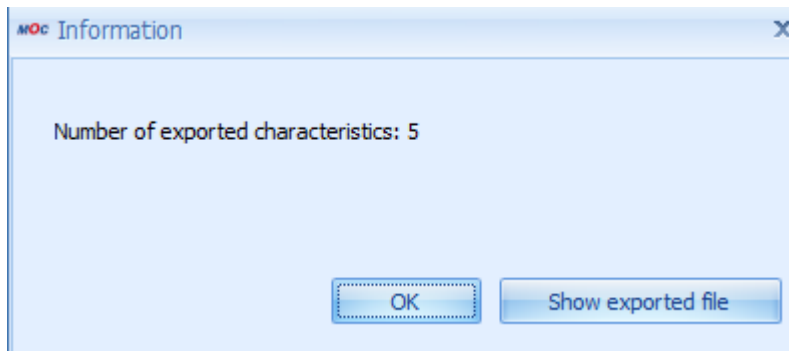
The data file generated is stored in the directory of the HYDRA path "QMIMP". When you import the characteristic data for an inspection plan later on, the data file for the import is selected in this path. If the HYDRA path "QMIMP" does not exist, you must create this path in the HYDRA path configuration.

The below screenshot shows a possible path configuration for the import of HYDRA-FMEA characteristic data.

You must not only specify the HYDRA path, but you must also ensure that the specified URL path is available on the HYDRA server. If the specified path is not available, you must manually create the path on the HYDRA server.

For the final import, you require the functions of the inspection planning based on CAD. For this reason, the above path configuration must be identical to the path configuration for the import of the CAD data file.

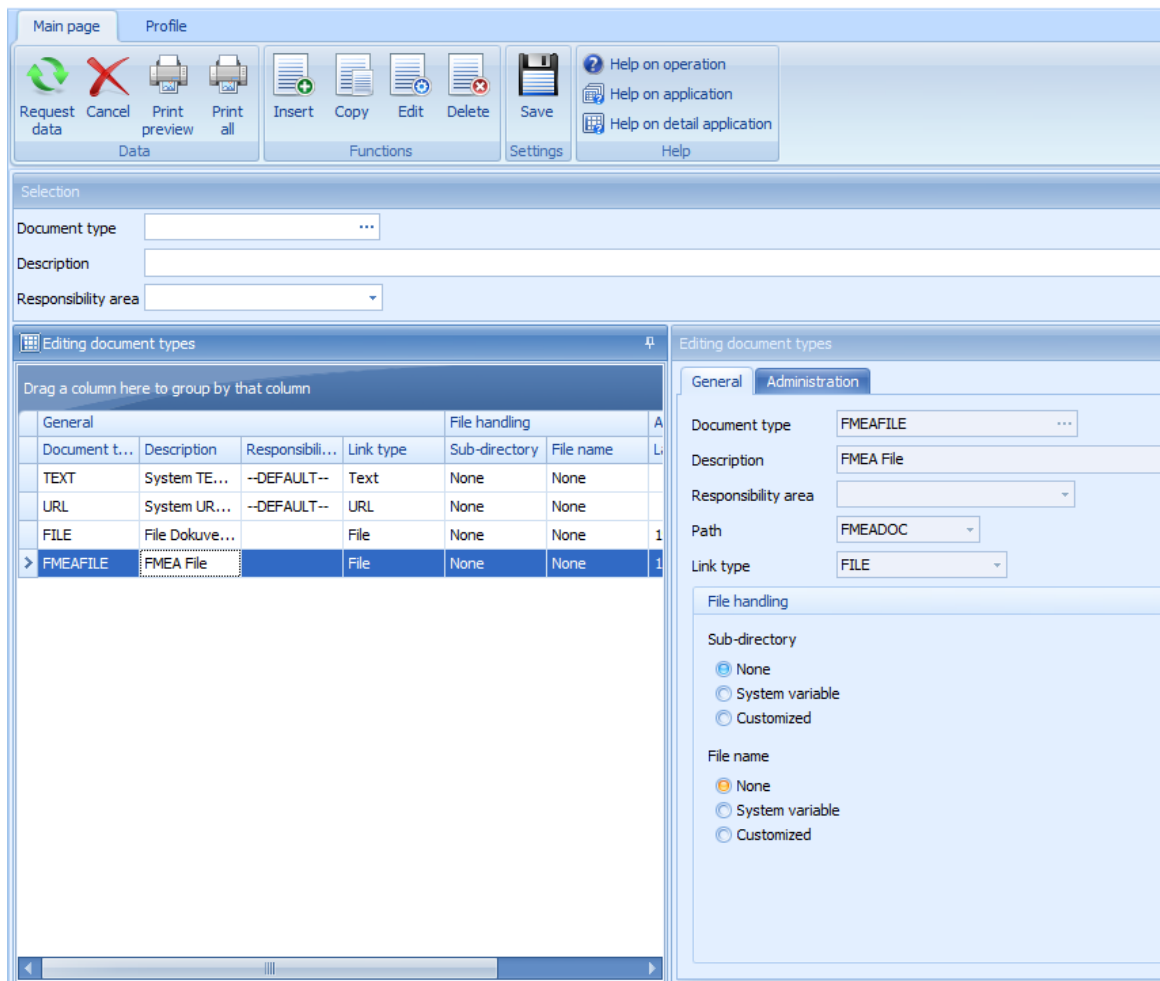
After the data export, the result is displayed in a dialog. You can also show the exported data file in the dialog.



## Attaching documents

To attach documents to an FMEA, the following configuration is required:

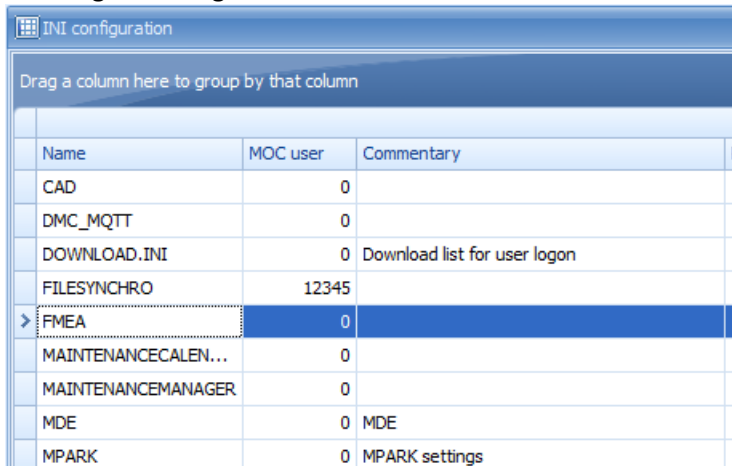
- Define own document types for the FMEA that can be stored in a separate location on the server.
- And: Create a new document type in the application *Editing document types*, e.g. FMEAFILE.



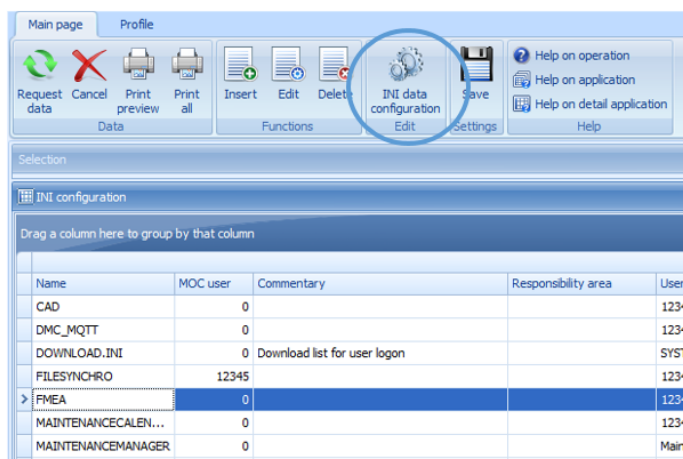
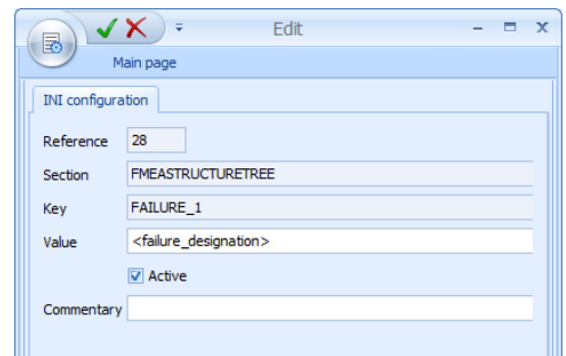
## Changing the label texts

Labels are dynamically created. You can specify the label layout. Create an INI configuration to this end.

Creating INI configuration:



| Name                | MOC user | Commentary                   |
|---------------------|----------|------------------------------|
| CAD                 | 0        |                              |
| DMC_MQTT            | 0        |                              |
| DOWNLOAD.INI        | 0        | Download list for user logon |
| FILESYNCHRO         | 12345    |                              |
| > FMEA              | 0        |                              |
| MAINTENANCECALEN... | 0        |                              |
| MAINTENANCEMANAGER  | 0        |                              |
| MDE                 | 0        | MDE                          |
| MPARK               | 0        | MPARK settings               |

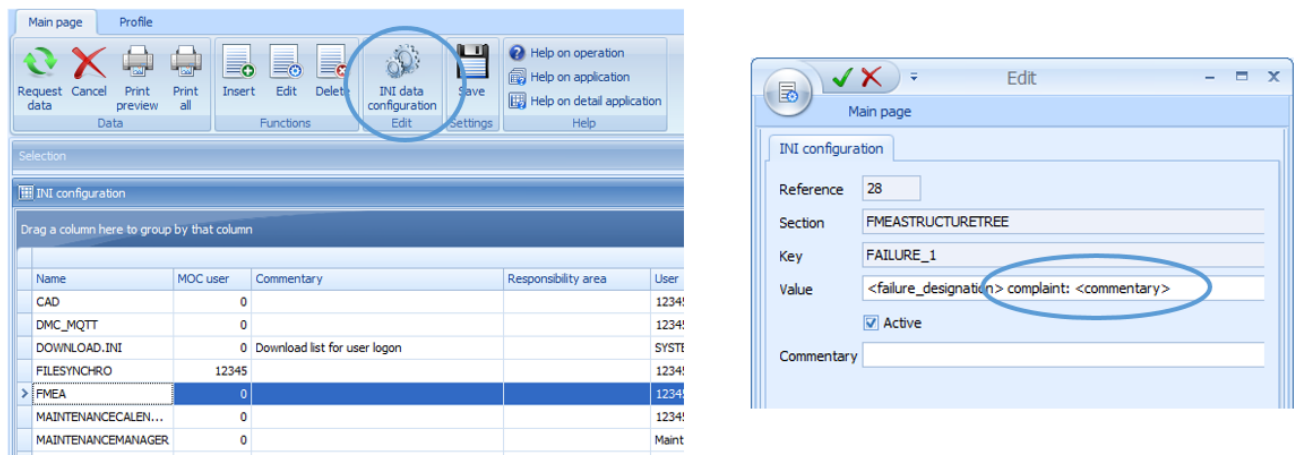
| INI configuration |                                     |
|-------------------|-------------------------------------|
| Reference         | 28                                  |
| Section           | FMEASTRUCTURETREE                   |
| Key               | FAILURE_1                           |
| Value             | <failure_designation>               |
| Active            | <input checked="" type="checkbox"/> |
| Commentary        |                                     |

Creating INI data configuration:

- Name: FMEA
- Create different labels for the structure element and for the network/network tooltip/tree, i.e. for the tree 7 different types, for network (label and tooltip) 3 different types each (failure, function, characteristic).
- You need not store definitions for all types, only for the types that overwrite the standard.
- Available sections:
  - FMEASTRUCTURETREE
  - FMEASTRUCTURENET
  - FMEASTRUCTURENETTOOLTIP
- Available keys
  - PROCESS
  - SYSTEM
  - FUNCTION
  - CHARACTERISTIC

- FAILURE
- MEASURELEVEL
- MEASURE
- For each INI configuration, you can configure 128 characters as value. To define longer labels, you can add an "\_n" to the keys with n being the number of the relevant entry.
- Up to 9 different entries are supported, from "\_1" to "\_9", i.e. a total of over 1100 characters.

Example process element:



- Specify the properties in angle brackets <acronym>.
- Note: If you use several INI data configurations for an element type, only save complete words, because a space character is added at the end of an INI data entry. It is NOT permitted:
  - PROCESS\_1 ending with "<acro"
  - PROCESS\_2 beginning with "nym>"
- The following fields are available for the label configuration:
  - All fields of list service if useful and WITHOUT fields with time without field gauge\_type
- Special features:
  - The fields
    - lower\_tolerance\_limit\_rel,
    - lower\_tolerance\_limit,
    - upper\_tolerance\_limit\_rel,
    - upper\_tolerance\_limit,
    - target\_value
 are formatted using the specified number of decimal places in the element (decimal\_places).
  - The fields
    - actual\_date
    - target\_date
    - measure\_date
 are formatted using a date format that is reproduced in the field. Use the following syntax in these fields: <acronym;DATEFORMAT> with DATEFORMAT providing the following options:
    - yyyy for a year with four digits

- MM for a month with two digits
- dd for a day with two day

Examples:

- The configuration <measure\_date;dd.MM.yyyy> is used to display the revision state in format 27.02.2013.
  - The configuration <measure\_date;MM-dd-yyyy> is used to display the revision state in format 02-27-2013.
- 
- Special feature of labels and tooltips in the network: Additionally to the fields of the actual element, you can use parent node fields of the system or process element. A "SYSPROCELEM" is added to these acronyms, e.g. <SYSPROCELEM.designation>.